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The Geometric Framework for Nutrition and Its Application to Rodent Models

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Keywords

appetite, lifespan, macronutrient, metabolism, nutrition, protein

Abstract

Rodents have been the primary model for mammalian nutritional physiology for decades. Despite an extensive body of literature, controversies remain around the effects of specific nutrients and total energy intake on several aspects of nutritional biology, even in this well-studied model. One approach that is helping to bring clarity to the field is the geometric framework for nutrition (GFN). The GFN is a multidimensional paradigm that can be used to conceptualize nutrition and nutritional effects, design experiments, and interpret results. To date, more than 30 publications have applied the GFN to data from rodent models of nutrition. Here we review the major conclusions from these studies. We pay particular attention to the effects of macronutrients on satiety, glucose metabolism, lifespan and the biology of aging, reproductive function, immune function, and the microbiome. We finish by highlighting several knowledge gaps that became evident upon reviewing this literature.

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INTRODUCTION

Failure to ingest a nutritionally balanced diet has profound consequences for animal health and performance, as well as downstream consequences that may affect group dynamics, trophic interactions, and ecosystems (1, 2). Attaining nutrient balance is no simple matter, however, requiring coordinated intakes of numerous macro- and micronutrients, which come co-packaged as complex mixtures within foods, often along with other substances (e.g., fiber, toxins, microbes). Those foods may themselves be scattered in time and space and be costly and dangerous to find, ingest, and digest. Furthermore, the multidimensional nutritional requirements of animals are not static, changing over the life course and with circumstance (3). Taming the complexity of nutrition therefore requires a highly integrative framework, one that embraces the dynamic, multidimensional nature of nutrition. Implementing such an approach to gain a better understanding of the consequences of nutrient intake also requires tractable and well-understood experimental systems.

Here we review the geometric framework for nutrition (GFN) and illustrate its implementation in rodent models, arguably the primary experimental system for nutritional physiology. The GFN provides a means to construct models that integrate diet, nutritional state, feeding behavior, physiology, and the consequences of nutrition across biological scales (from gene expression to life history outcomes). The method has been applied to approximately 50 taxa ranging from slime molds to invertebrates, fish, and numerous mammals, including humans (for reviews, see 1, 4–8 and also 9 in this volume). The underpinning theory and progressive developments in the implementation of the GFN have been described many times since its inception in the early 1990s (1, 3, 7, 10, 11). Our specific aim is to discuss nutritional and pathophysiological uncertainties that have been resolved by applying the GFN to rodents, one of the most widely used models in nutritional biology; see also Reference 9 for some applications to the most widely used nonhuman primate biomedical model, the rhesus macaque. We begin by covering the core concepts of the GFN as well as experimental design and analysis.

The Nutrient Space and Intake Targets

The GFN is a state-space model oriented around a multidimensional nutritional space. The nutrient space is n-dimensional (hypothetically including dimensions for all micro- and macronutrients and other substances of interest), though 2D spaces convey the principles readily. A 2D nutrient space is represented as Cartesian space, where the x- and y-axes denote amounts of two nutrients, such as protein and carbohydrates, and coordinates within the space represent intake amounts (**Figure 1*a***). Foods are represented as vectors (food rails) radiating from the origin with a slope equal to the food's ratio of nutrients (**Figure 1*a***). An intake target (IT) gives the location within the space that represents the nutritional state that an organism's appetite seeks to attain, i.e., the amount and ratio of nutrients consumed if given a free choice of foods. Individual foods may be imbalanced with respect to the IT, but where several foods are nutritionally complementary, an animal may selectively consume from them to reach the IT (e.g., **Figure 1*b***). In choice experiments, animals are subjected to two or more complementary foods, with their nutrient intake tracked to define the IT. A central premise of the GFN is that an IT is not fixed across life and may change with factors such as life stage, physiological state, and environmental conditions (reviewed in 1, 3). For example, GFN studies in wild primates have revealed how the demands of falling temperature and lactation affect the regulated intake of different energy sources (see Reference 9 in this volume). Below, we give examples from GFN studies in rodents where the location of the IT can be affected by extended periods of prefeeding on imbalanced foods (12), or even by microbiome composition (13).

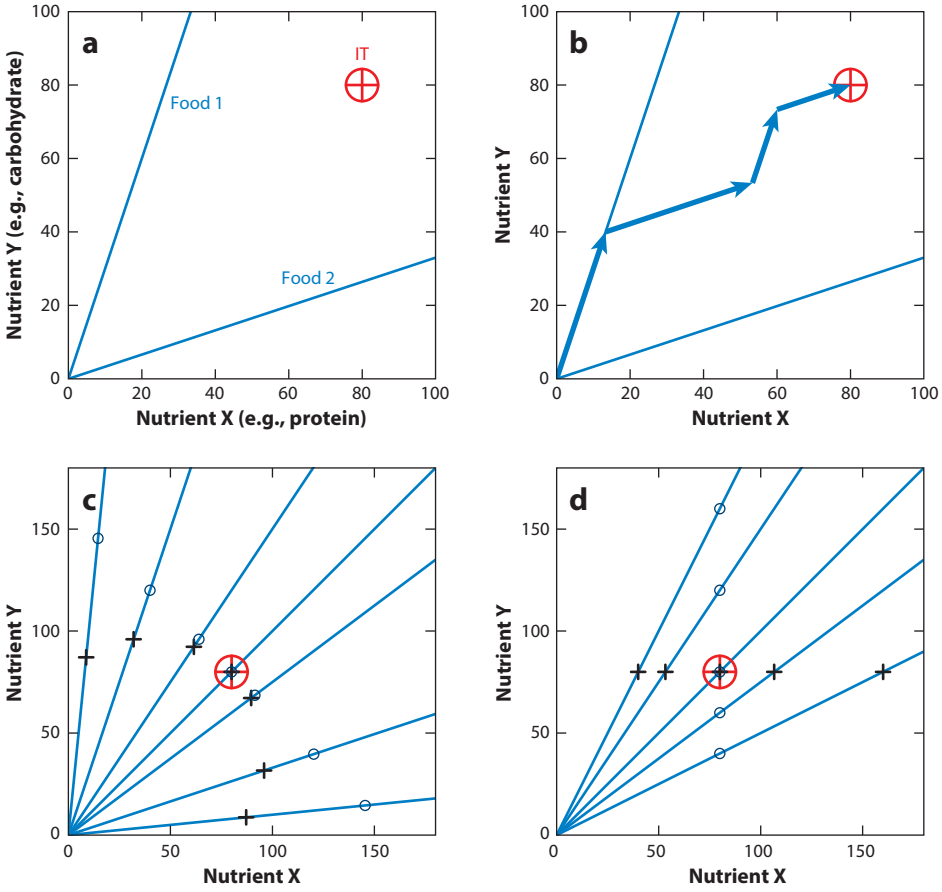

Figure 1

Core concepts of the geometric framework for nutrition. (*a*) A 2D nutrient space where intake of one nutrient (e.g., protein, P) is shown on the *x*-axis and another (e.g., carbohydrate, C) is shown on the *y*-axis. Foods are shown as rails (*solid lines*); here, food 1 has little P compared to C (*steep slope*), whereas food 2 has a higher P:C ratio (*shallower slope*). The intake target (IT; *red cross*) shows the location an animal's appetite system will guide it to under free choice. (*b*) Example of a choice experiment in which individual foods are imbalanced with respect to the IT (i.e., the rail does not pass through the IT), but the IT falls within the space encompassed by the foods, making them complementary. Given a choice to feed freely, an organism can find a route to the IT (e.g., *bold arrows*). (*c*,*d*) Examples of nutritional compromises animals may make in no-choice experiments where only one food is available. (*c*) The equal distance rule (*open points*) and closest distance rule (*crosses*). (*d*) Defend the intake of the nutrient X (*open points*) or nutrient Y (*crosses*).

In no-choice experiments, groups are offered ad libitum access to one food, locking each group onto a rail with intake tracked. The pattern of intake observed from groups on different foods informs the compromises that an organism makes when faced with imbalanced foods. Among infinite potential rules of compromise, four patterns are illustrative. A linear array of points that are equidistant from the origin (**Figure 1*c***), termed the equal distance rule, indicates that the organism has maintained net intake of nutrients equal to the total at the IT [e.g., regulating total energy intake regardless of macronutrient source, as recorded for a wild population of rhesus macaques (9)]. Another pattern is an arc-shaped array, referred to as the closest-distance rule,

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which minimizes distance to the IT (**Figure 1*c***). Alternatively, the organism may defend its intake of one nutrient at the expense of under-/overconsuming the other (**Figure 1*d***).

Response Surfaces and Consequences of Nutritional Imbalance

GFN experiments have been used to disentangle the long-term dietary effects of macronutrients and energy on outcomes of interest. Such studies lock animals onto food rails for extended periods and may even experimentally disperse groups along those rails, thus separating the effects of amounts and ratios of nutrients eaten (**Figure 2*a***). To date, GFN studies have been concerned with ad libitum food intake, so dispersal along rails has been done by diluting the nutrients within the foods, achieved by adding indigestible cellulose to pelleted foods (14) or water to liquid-based diets (15). The choice of dilutant is potentially important and may have to be considered in the interpretation of the experiment. Theoretically, organisms could be dispersed across the nutrient space by limiting food access or by forced overfeeding. In the case of the former, the experiment would potentially capture effects of fasting; animals are likely to consume their aliquot of food within a short period, being food deprived for the remainder of the day [as seen in studies of caloric restriction (CR) (16)].

Once responses on each diet have been measured, they can be mapped to the nutrient space. **Figure 2*b*** shows an example response surface that came from the diets in **Figure 2*a***. Here the gradient of response to nutrient X is steeper than for nutrient Y; the food with the highest density of nutrient X (the purple rail in the figure) crosses more contours than the food with the higher nutrient Y (the green rail). Where the hypothesis being tested relates to nutrient intake, as opposed to diet composition, it can be informative to recreate surfaces as a function of nutrient intakes; this is because different diet compositions may be more or less satiating (e.g., 17). For example, the surface on **Figure 2*b*** may depict dry-weight food intake, indicating that nutrient X has a satiating effect. In such a case, other responses of interest (e.g., lean mass) might be recast over the nutrient space as in **Figure 2*c*** to depict effects of nutrient ingestion. In the example in **Figure 2*c***, isocalorically substituting the intake of X for Y (i.e., across the dashed line) increases the outcome. Just as the IT may be affected by life stage, physiological state, and/or the environment, the shape of a nutrient surface may also change with such factors. For example, the response of lifespan to dietary macronutrients in *Drosophila melanogaster* likely varies with temperature (18). Below we give examples from GFN studies in mice demonstrating how dietary macronutrients interact with life stage to affect mortality (19) and with endocrine disorders to affect reproductive function (20).

Three-Dimensional Experiments

The GFN has been applied in 3D with an emphasis on separating the effects of protein (P), carbohydrate (C), and fat (F). The concepts of nutrient space, IT, food rails, rules of compromise, and response surfaces discussed above translate to 3D. However, adding a third dimension (or more) adds practical complexities to experimental design and interpretation. Most prominently, many diet formulations are required to factorially test compositions in 3D.

Regarding visualization, compositions in 3D have often been depicted on a simplex. **Figure 2*d*** provides an example simplex showing 10 foods sampling compositional space with % energy from P between 5% and 60% and from C and F between 20% and 75%. Simplexes can be challenging to read, and the GFN's main contribution to mixture experiments has been to propose the right-angle mixture triangle (RMT), which is more intuitively interpretable (21). On an RMT, two proportions are plotted in an x, y space bounded at 0 and 1, whereas the third proportion is on an implicit axis, which increases in concentration from the hypotenuse toward the origin. **Figure 2*e*** casts the same diet compositions from **Figure 2*d*** into RMT space. Though easy to read, RMTs

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cause some extension of distances along the implicit axis, meaning that effects associated with the implicit nutrient need to be interrogated closely for magnitude. See Reference 9 for an example of the application of RMTs in an ecological context.

An additional issue when studying macronutrients is that F has an energy density roughly double that of P and C. Hence, to control energy density of foods (kJ/g), an indigestible dilutant

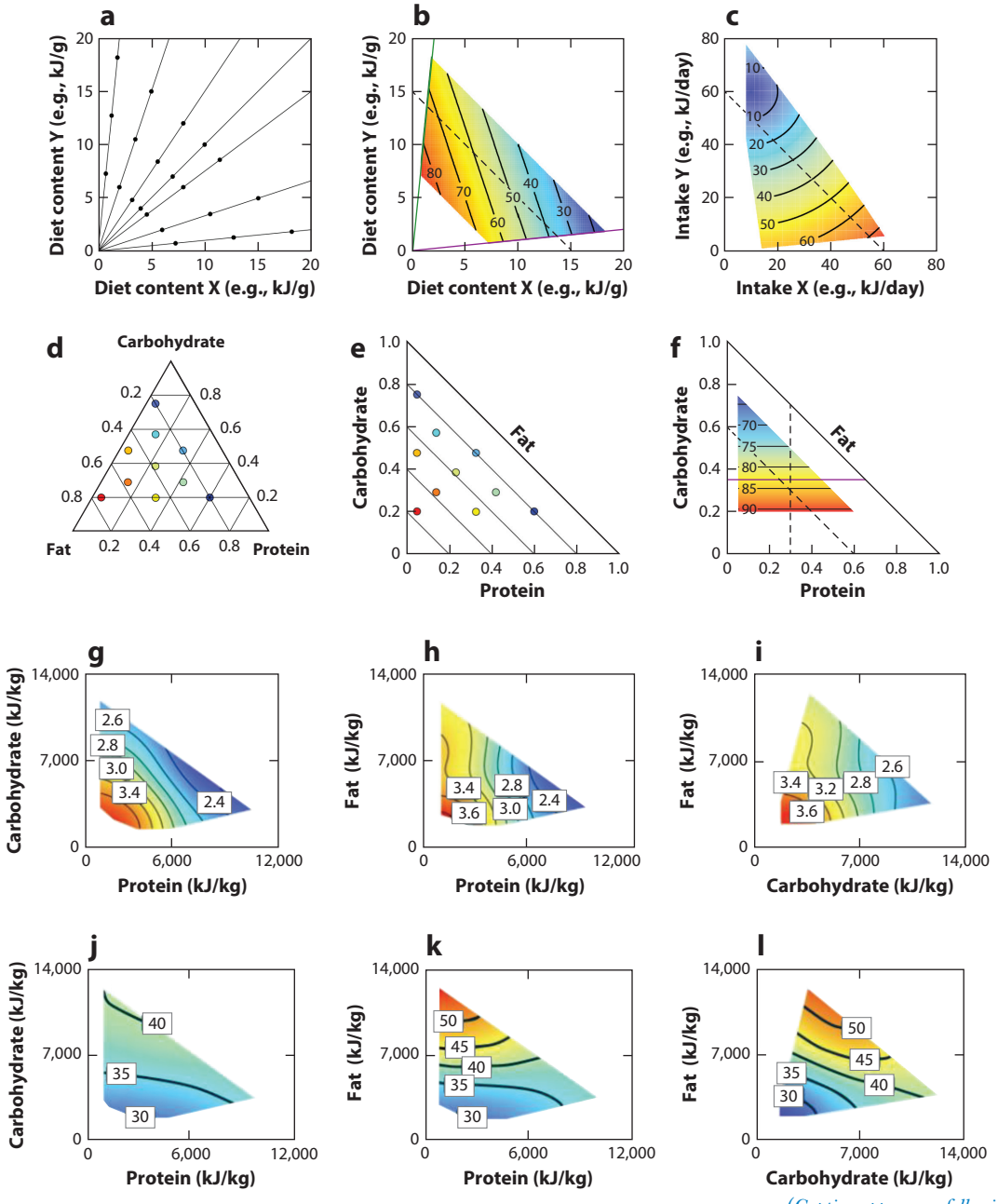

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Figure 2 (*Figure appears on preceding page*)

The design and interpretation of geometric framework for nutrition (GFN) experiments. (*a*) A GFN experiment designed to separate the effects of nutrient ratio/diet composition and total nutrient intake. Here diets are composed of 7 nutrient ratios (i.e., 7 food rails) replicated at 3 densities. (*b*) Example of a response surface displaying some outcome as a function of the diets in panel *a*. Surfaces are predictions from regression-based statistical models fitted from the experimental data and are read like a topographical map (red = high values of response, blue = low values, contours depict elevation). The dashed line overlaid on panel *b* is an example of a substitution vector, where equal parts X are substituted for Y. (*c*) Example of a response surface fitted from intake data rather than diet content. (*d*) Example of a simplex showing 10 diets that systematically vary in their contents of protein, carbohydrate, and fat (for reference, higher-fat diets are shown in warmer colors). (*e*) Example of a right-angle mixture triangle (RMT) depicting the same diets as in panel *d*. (*f*) A response surface fitted over an RMT showing how diet composition in 3D affects an outcome. (*g–i*) Food intake (g/mouse/day) and (*j–l*) energy intake (kJ/mouse/day) surfaces in C57Bl/6J mice as a function of (*g,j*) protein and carbohydrate, (*h,k*) protein and fat, and (*i,l*) carbohydrate and fat content of the diet. Panels *g–l* created using data from Reference 14.

must be added that covaries with F% (e.g., the warmer-colored foods in **Figure 2*d,e*** may have the highest cellulose concentration). The alternative is to hold the indigestible fraction of the diet at a constant dry weight, though in this case increasing F% would be associated with increasing energy density. One may include both diluted and nondiluted series of diets in a study, then use independent (e.g., biological) criteria to decide whether the response is driven by F, energy content, or the dilutant (e.g., cellulose).

Because rodent experiments are temporally and economically costly, many 3D GFN studies have held energy density constant. That is, they lock groups on food rails (e.g., the compositions in **Figure 2*d,e***) but do not disperse the subjects along them (except through natural variation in food intake within the experimental cohort). In this case, the response to the experiment can be visualized as a surface on a simplex or an RMT. For example, **Figure 2*f*** shows a response surface, with an effect driven predominantly by C; energy from C can be substituted for either F or P, and contours are crossed (dashed lines), but substituting F for P (holding C constant) crosses no contour (purple line).

THE GEOMETRIC FRAMEWORK FOR NUTRITION IN RODENTS

Food and Nutrient Intake

The first GFN application to rodents was Simpson & Raubenheimer's (12) reanalysis of feeding experiments in rats. They began with data from choice experiments in adult male Sprague–Dawley rats (22). The rats were given a series of isocaloric food pairs that varied in the ratio of P:C:F, with intake measured. Simpson & Raubenheimer (12) recast the intake data in nutritional space, showing that the amount and ratio of macronutrients consumed were defended at an approximate P:C:F ratio of 5:13:5 g/day (a value that differed from random selection of the available foods), giving an estimate of the IT.

Simpson & Raubenheimer (12) then evaluated how nutritional history affects intake and the rules of nutritional compromise. They reanalyzed data from young male Sprague–Dawley rats prefed with excessively high/low P:C (23). The intake of animals in experiments where the IT could and could not be reached was then analyzed. Given free rein, animals preferentially consumed large amounts of the formerly deficient macronutrient. However, when faced with further nutritional compromises, low-P (LP) animals still strictly defended P intake, even when suffering large deficits/excesses of C.

The first rodent experiment to be conceptualized with the GFN was in 5-week-old male NMRI mice (24). Sørensen et al. (24) first undertook choice trials that tested foods with different P:C ratios (casein versus starch; fixed F). The animals consumed the foods nonrandomly, converging on similar points in nutritional space across trials, indicating nutrient-specific regulation of both P and C. The estimated IT was P:C:F% energy 24:58:18%. Over the course of a 32-day

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no-choice experiment where animals were confined to one of five foods with between 9% and 48% P (g/100 g) the animals' intake formed a pattern that fell between the equal distance rule (open points in **Figure 1*c***) and protein defense (open points in **Figure 1*d***). Hence, on diets with a P:C lower than the IT, animals overconsume C more than they under-consume P, and vice versa. A second GFN study by Huang et al. (25) found the same pattern of nutrient regulation in male C57Bl/6J mice.

Protein Leverage and FGF21

The experiments by Sørensen et al. (24) and Huang et al. (25) suggest that under laboratory conditions male mice possess an appetite that regulates total P intake more strongly than non-P intake. This appetite means that the animals are willing to eat a relative excess of non-P intake on LP diets or a deficit of non-P intake on high-P (HP) diets. A consequence of this form of nutrient regulation is that as the density of P relative to other macronutrients in the diet falls, total food and energy intake rise, up to a point of protein dilution below which the response fails and protein malnutrition occurs (24–26). This phenomenon has been termed protein leverage and has been demonstrated in a range of other species, including humans, where it has been implicated as a potential integrating mechanism in obesity (6, 27, 28).

Several GFN experiments, along with numerous non-GFN studies (e.g., 29–32), have now provided evidence for protein leverage in mice (14, 20, 33), albeit not as strongly as in other species, including humans (6). Solon-Biet et al. (14) reported the most comprehensive of these studies, undertaken in C57Bl/6J mice of both sexes from weaning. The study involved pelleted foods containing the 10 P:C:F ratios in **Figure 2*d***. Each composition was replicated at 8, 13, and 17 kJ/g, giving 30 diets, of which 25 sustained growth [failing diets were lowest (5%) P at low or mid-energy]. Energy density was controlled by adding indigestible cellulose. This study clearly showed that P intake was prioritized over C and F intakes. Here we have redrawn the key findings around food and energy intake and dietary macronutrients in **Figure 2*g–l***. Isocaloric substitution of P for either C or F in the diet led to strong reductions in dry-weight food intake (**Figure 2*g–i***), with consequentially lower energy intake (**Figure 2*j–l***).

Crean et al. (34) and Rodriguez Paris et al. (20) also reported the effects of dietary macronutrient composition on energy intake in male and female C57Bl/6J mice, respectively. Both studies used an isocaloric design, where the ratios of macronutrients varied but energy density was fixed. The results of the two separate experiments were strongly concordant. Food intake fell sharply up to approximately 20% P before plateauing. In an aligned result, Hew et al. (35) reported the energy intake of animals in a GFN study on wound healing in male mice (for further GFN studies on wound healing, see 36, 37). Here, energy intake was substantially elevated in animals on a 5% P diet relative to 14% or 16% P.

There has been widespread interest in the mechanisms underlying regulation of protein intake, with a focus on the hormone fibroblast growth factor 21 (FGF21). Many studies have shown that mice fed on LP diets have increased circulating levels of FGF21 (e.g., 38–42). However, there has been debate as to whether elevated FGF21 levels are the result of low P state, high C state, or low total energy intake, or perhaps even excited thermogenic pathways, resulting in elevated energy expenditure that secondarily elicits compensatory feeding (17, 38, 43).

The GFN has helped resolve these contrasting claims. Solon-Biet et al. (44) reported circulating levels of FGF21, as well as hepatic messenger RNA expression of *FGF21*, in 15-month-old animals coming from a large 25-diet GFN study (7). Modeling these outcomes as nutrient surfaces demonstrated that elevated FGF21 is associated with low P intake, independently of energy intake. More recently, GFN study designs have been applied to mice receiving endogenous FGF21

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infusion, as well as *FGF21* knockouts (45). In choice studies offering foods with variable P:C ratios, infusion of FGF21 robustly resulted in elevated P intake but did not affect C intake. Solon-Biet et al. (45) also confirmed previous results that FGF21 affects sucrose preference and energy expenditure, but that these effects depend on carbohydrate and energy status, respectively.

It is worth contrasting the GFN studies that detect evidence for protein leverage with a study by Hu et al. (46) that purported to find no evidence for protein leverage. Like Solon-Biet et al. (14), Hu et al. (46) studied the effects of macronutrient composition on food intake in C57Bl/6J mice on 29 diets (analyzed over a 4 series of experiments). Unlike Solon-Biet et al. (14), Hu et al. (46) report no statistically significant evidence for protein leverage but rather conclude that dietary F drives elevated energy intake via a hedonic effect. The discrepancy in conclusions between the two studies can almost certainly be resolved by the application of the GFN; such an analysis allows for partitioning the effects of the different macronutrients captured within the whole study, the elevated energy density of high-F diets, and the nonlinear signature of P leverage.

Glucose Metabolism and Insulin Resistance

Elevated energy intake driven by protein leverage has been observed to increase body mass and adiposity in the studies described above (24–26, 33). Other correlated morphological and metabolic effects have also been noted. Sørensen et al. (26) showed that increasing P:C ratio and concomitant decreases in food intake result in increased stomach mass and decreased intestine, ceca, and colon masses. Huang et al. (25) showed that the increased energy intake driven by protein leverage coincides with increased body temperature, an effect that could be traced to gene expression in brown adipose tissue. An obvious further question is how macronutrient content relates to metabolic health and insulin resistance. We now review the insights into this question that have been derived from the GFN.

Several studies have used GFN to investigate the effects of macronutrients on glucose metabolism. Solon-Biet et al. (14) report the effects of P, C, and F on glucose metabolism in 15-month-old mice. They measured many outcomes, including circulating levels of insulin, area under the curve (AUC) in a glucose tolerance test, liver triglycerides, and more. LP diets resulted in elevated energy intake and adiposity (i.e., protein leverage). Despite weight gain, animals with elevated C and low P intakes had the best late-life parameters of metabolic health. In contrast, high P intake was associated with poorer outcomes, despite the animals being leaner. Similar negative effects of higher P intake on parameters such as fasting blood glucose have been seen in at least three other GFN studies in C57Bl/6J mice (20, 34, 47).

Further GFN support for the negative effects of dietary P on glucose metabolism comes from a meta-analysis of studies that manipulated dietary branched-chain amino acid (BCAA) content (48). In this analysis, the underlying studies used a pairwise comparison of high- versus low-BCAA diets. The authors then separated the effects of manipulating BCAAs versus non-BCAAs via the GFN. The results show that simultaneous manipulation of both BCAAs and non-BCAAs drives the biggest increases in glucose AUC, suggesting, by this measure at least, that high total P is detrimental. Furthermore, Solon-Biet et al. (49) explored the effect of the ratio of BCAA to non-BCAA (holding total P constant) in 312 male and female C57Bl/6J mice. They showed that elevated BCAA:non-BCAA was associated with hyperphagia and obesity, with tryptophan being a key non-BCAA. These data suggested a double burden for elevated BCAA—both as part of overall P content in the diet and as a driver of hyperphagia via interactions with other amino acids such as tryptophan.

High-fat, energy-dense Western diets robustly result in insulin resistance, particularly in C57Bl/6J mice (50–52). The insights from the GFN discussed here strongly suggest that these

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effects are likely to be driven by diet energy density, rather than fat being toxic per se. That is not to say that dietary fats have no metabolic effects independent of calories, just that these effects are likely tissue specific. For example, Farris et al. (53) present the results from RNA sequencing applied to the fat pads of males from an isocaloric 10-diet GFN study. Here genes were clustered by response to macronutrients. The largest clusters were dominated by F effects. The cluster that was positively associated with F was enriched for genes associated with ciliary function and the autosomal recessive ciliopathy, Bardet–Biedl syndrome, which imparts symptoms that include obesity (54).

The GFN studies discussed above used sucrose and starch as their C content. However, a pair of GFN studies by Wali and colleagues (42, 55) has interrogated different C sources. In both cases, the experiments involved glucose tolerance tests of male C57Bl/6J mice after 6 weeks on a diet. The studies used an isocaloric design (total energy = 14.5 kJ/g). The amount and composition of C were varied by manipulating glucose and fructose, sucrose, and native and resistant starch across 37 treatments at fixed F, whereas P covaried with total C (42). In the second study, P was fixed and F varied, in both concentration and type (soy oil versus lard—a distinction that made no difference to outcomes measured) (55). A key feature of both experiments was the titration of fructose and glucose. Food intake and adiposity peaked at a 1:1 mix of glucose to fructose, an effect that was further exacerbated by P leverage on LP diets (42). Even correcting for total energy intake, the 1:1 mix drove elevated insulin resistance and liver triglycerides. The effect was observed regardless of F and P content (42, 55). Wali et al. (42) separated the effects of sucrose and wheat starch but did not observe strong detrimental effects of either. Given that sucrose is a disaccharide composed of glucose and fructose, the contrasting results of the two experiments suggest that a 1:1 monosaccharide mix of glucose and fructose is particularly harmful. Interestingly, this mix is found in high-fructose corn syrup (56). By contrast to native starch and sucrose, inclusion of resistant starch in the diet promoted markers of metabolic health and was associated with a distinct gut microbiota.

The studies by Wali and colleagues helped resolve the protein paradox—the apparently contradictory observations that LP, high-C diets drive excess energy intake via protein leverage, while simultaneously imparting insulin tolerance and, as we explore in the next section, sustaining longest life. Wali et al.'s work showed that a key to resolving this paradox lies in the type of C: a LP, high-C diet containing complex C promoted the best metabolic outcomes, whereas an equivalently LP diet high in glucose and fructose provided the worst outcomes.

Aging

The rodent has been a major model for understanding the effects of diet on lifespan and the underlying biological processes that cause aging itself. Several studies have now used the GFN to show how specific macronutrients affect lifespan and aging in rodents. We begin by reviewing the effects of macronutrients on rodent lifespan, as detected via the GFN, then delve into the underlying mechanisms.

Lifespan. Many studies in rodents have demonstrated that CR treatments, which reduce total calorie intake by ~30% without malnutrition, can result in extended lifespan (57, 58; see also Reference 9 in this volume for discussion in primates). The lifespan-enhancing effects of CR have been attributed variably to a reduction in net calories, reduced intake of specific nutrients (e.g., protein or amino acids), or periods of fasting (16, 59–62). One of the main aims of the large 25-diet GFN study covarying P:C:F composition and total energy density (first described in 14) was to separate the effects of energy intake and source on lifespan within an ad libitum context and thereby resolve this controversy.

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Solon-Biet et al. (14) modeled the median lifespan on each diet as a function of macronutrient intake. The lowest-energy diets, which had levels of energy intake comparable to CR, had low lifespan. Median lifespan was maximized with LP, high-C intakes. Sex-stratified GFN analyses of lifespan were presented in a separate publication showing the same trend; P:C ratios of 1:13 and 1:11 maximized lifespan in males and females, respectively (63). Senior et al. (19) undertook a secondary analysis of these data to look at age-specific mortality. The first part of these analyses concorded with those presented previously: LP, high-C diets result in maximal life expectancy by minimizing mortality through middle life. However, for animals that reached older age (>100 weeks), mortality was minimized on higher P:C diets (19), suggesting a potential benefit of higher protein in older mice.

Speakman et al. (60) quantitatively compared rodent studies using conventional CR versus those with CR without protein restriction. While the constituent studies did not use GFN designs, Speakman et al. (60) presented the observed effects using a GFN-derived approach. Their work suggests that although protein restriction results in lifespan extension, a larger effect is attributable to reduced total energy. A key distinction between the studies assessed by Speakman et al. (60) and the experiment in Solon-Biet et al. (14) is that the latter allowed ad libitum food, but the former enforced CR through restricted access to food with attendant periods of imposed fasting (e.g., 16).

The biology of aging. The GFN has also been applied to understand the effects of macronutrients on markers of aging biology and age-related health in mice. Fotheringham et al. (64) reported on the renal health of the aged animals. High P, low C intakes maximized kidney weight and minimized proinflammatory cytokines and markers of kidney injury. The authors highlight, however, that these effects fall within a range that was considered prepathological (64). Notably, the effects of macronutrient intake on lifespan and metabolic health are likely to be inversely correlated with the effects on renal function in old age.

Changes in skin structure and dentition are one of the most visible signs of senescence in mammals. The effect of macronutrients on skin structure and alveolar bone loss in aged mice were outcomes from Solon-Biet et al.'s (14) large GFN study (65, 66). In both sexes, subcutaneous skin was thickened by increasing C and F intake with little effect of P. In contrast, high P, low C intake maximized dermal skin thickness. Macronutrient intake had sex-specific effects on epidermal skin thickness, with males benefitting from higher P and females from high C intakes (65). High C intake was correlated with the largest degree of alveolar bone loss, regardless of whether C intake was in the form of sucrose or starch. The intake of C also correlated positively with inflammatory markers in gingival tissue, though the effect was not statistically significant (66).

Of all the organs in mice from the large GFN lifespan study, the aging liver was studied most extensively. Cogger et al. (67) reported on the effects of macronutrient intake on age-related liver defenestration. Fenestrations are trans-cellular pores perforating the liver sinusoidal endothelial cell and support liver function by facilitating transfer between blood and hepatocytes (68). Age-related reductions in fenestration size and porosity likely contribute to impaired hepatic clearance in older age (69). Cogger et al. (67) showed that moderate F intakes maximized fenestration porosity in older animals. However, both C and P have positive and additive effects on defenestration frequency. By correlating individual animal responses, Cogger et al. (67) implicated the microbiome and circulating levels of free fatty acids with fenestration parameters, providing potential mechanisms by which diet may affect liver function in old age.

The effects of diet on hepatic telomere length were also studied in the 15-month-old animals (70). Telomeres are the caps on the ends of eukaryotic chromosomes, comprising stretches of repeating DNA sequence. Telomeres shorten with mitotic divisions and chronological age, and shorter telomeres have been implicated as a biomarker of, and causal factor in, aging (71). The

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GFN study showed that at 15 months of age, LP, high-C diets were associated with the longest telomeres and that telomere length declined with increasing P:C ratio. The GFN surfaces for telomere length correlated strongly with the surfaces for median lifespan (70).

As well as the effects on telomeres, hepatic gene expression in 15-month-old culled animals from the large GFN study has also been evaluated (14, 72). Solon-Biet et al. (14) show that hepatic *mechanistic target of rapamycin* (*Mtor*) activation is significantly reduced with high C, low P intakes, coinciding with maximal lifespan; mTOR is a central regulator of multiple pathways implicated in lifespan and has been shown to respond to amino acid status in other contexts (73). Gokarn et al. (72) undertook a more systematic analysis of late-life hepatic gene expression in these animals. They found that the expression of more genes was affected by P than by C or F intake. P also affected many genes previously implicated in aging biology (e.g., *Mtor*, *insulin-like growth factor 1*, *nicotinamide N-methyltransferase*, and *Fgf21*). Gene ontology analysis identified key pathways and processes affected by P intake, which included mitochondrial function and, unsurprisingly, metabolic signaling and metabolism of amino acids (72).

A second experiment used the GFN to look at gene expression and aging in the liver. Le Couteur et al. (47) performed a large GFN experiment in mice involving 40 treatment groups. The nutritional design included 5 P:C:F compositions each at 2 energy densities. Each of these 10 diet groups was then replicated with 4 drug treatments: control (no drug), metformin, rapamycin, and resveratrol. These drugs have all been implicated previously in the biology of aging and interact with nutrient-sensing pathways. Hence, this study was designed to understand the interactive effects of drugs and diet on aging biology. Male C57Bl/6J mice were held on the treatments from 12 to 40 weeks of age with ad libitum food access before being culled. Liver samples were then assayed by proteomics.

This experiment validated the previous GFN study on liver gene expression. The results again found that P intake affected the expression of many liver proteins involved in mitochondrial function and amino acid metabolism (47, 72). There was evidence that high P intake was also associated with elevated oxidative stress, an effect that was somewhat alleviated by metformin and resveratrol, implying a geroprotective effect of the drugs. However, the major dietary dimension of the effect was total energy intake. Energy intake affected approximately 2,000 proteins. Proteins with expression positively affected by energy intake were involved in nutrient-metabolizing pathways, including glycolysis, gluconeogenesis, lipogenesis, ketogenesis, and many others. The subset of proteins that were negatively correlated with energy were enriched for representatives from the spliceosome and RNA metabolism. Similar effects have been seen in studies on mice using conventional CR by limiting food access (74). This concordant observation is not entirely unexpected given that the energy dilution reduced energy intake by ~30%, comparable to conventional CR. Treatment with the drugs generally dampened, and in some cases reversed, the responses of the liver proteins to intake of energy and individual macronutrients. Of the drugs, rapamycin and resveratrol impacted the relationship between expression and energy intake for more proteins than metformin (47). These drugs are often considered mimetics of CR. However, within this study, there was little evidence that they increased the abundance of proteins previously linked to CR. A possible explanation for this is that these drugs mimic fasting effects, which were not apparent in the current study where energy intake was achieved in an ad libitum setting.

Reproductive Function and Intergenerational Effects

Diet is known to affect reproductive health and function. Many studies have used the rodent as a model to attempt to understand how total dietary energy and the source of that energy might alter reproductive processes. Here we review how the GFN has been coupled with rodent models to understand the effects of diet on both male and female reproductive function. Importantly,

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GFN studies in rodents have also looked beyond the direct effects to consider how maternal and paternal diet might alter offspring physiology.

Female reproductive function. Female rodent reproductive function has been studied using the GFN in two contexts: in aged mice coming from the large GFN experiment described above (14, 63) and in a model of polycystic ovary syndrome (PCOS) (20). Solon-Biet et al. (63) reported on the effects of macronutrients on uterine mass, the number of follicles per ovary, the number of corpora lutea per ovary, and the number of estrous cycles within 11 days in 15-month-old C57Bl/6J mice.

Macronutrient intake affected all outcomes, with the unifying trend that low total energy intake was detrimental (63). However, the different traits were maximized at variable locations in the nutrient space. All three macronutrients positively affected uterine mass, with mass maximized on high intake (~20 kJ/day) of equal parts P:C:F. Follicle number was affected marginally by P intake and negatively by C intake and benefited from F intake. The number of corpora lutea and number of cycles negatively correlated with follicle number across the nutrient space and were affected positively by C and negatively by F. The authors concluded that in one respect the study aligns with research on CR demonstrating that low energy intake is associated with reproductive dysfunction (e.g., 75). They highlighted that the target intake of macronutrients that maximizes reproductive function for some traits (uterine mass and follicle number) in females is not aligned with that maximizing lifespan. Similar nutritional trade-offs between female lifespan and reproductive function have been observed in invertebrate models (e.g., 15).

The second study applied the GFN to a model of PCOS, an endocrine disease in women characterized by hyperandrogenism. PCOS can result in weight gain, insulin resistance, type 2 diabetes, and infertility (76). Treatments for PCOS focus on lifestyle interventions, including changes in diet, that reduce weight in the hope of alleviating other symptoms (77). Rodriguez Paris et al. (20) used the GFN to test for effects of dietary macronutrients on symptoms of PCOS in mice. Their experiment included 3-week-old female C57Bl/6J mice. One set of mice received a dihydrotestosterone implant that induced symptoms of PCOS, whereas the second had a sham control. The mice were then assigned to 1 of 10 diets, fixing energy at ~15 kJ/g but varying macronutrient composition in terms of % energy (P = 5–60%, C and F = 20–75%). Animals were given free access to the diets for 10 weeks, with body weight, adiposity, and food intake tracked and the number of estrous cycles over 11 days measured. Various circulating levels of hormones were assessed, as were markers of glucose metabolism. Post-cull, ovary morphology and the number of corpora lutea per ovary were measured.

Rodriguez Paris et al. (20) reported that the animals within the two groups exhibited similar patterns of food intake. Nonetheless, the PCOS treatment clearly induced dyslipidemia, elevated body weight and adiposity, and poorer markers of glucose metabolism (20). Diet could not alleviate the metabolic symptoms of PCOS treatment. Relative to the control animals, the PCOS animals displayed marked ovulatory dysfunction. However, a subset of animals did recover presence of corpora lutea and completed estrous cycles. These animals were confined to a small region of the nutrient space comprising 20–30 kJ/day of C, 15–25 kJ/day of F, and moderate P (~8 kJ/day). The authors concluded that macronutrient-based interventions could help alleviate the reproductive, but not the metabolic, impairments associated with PCOS. Macronutrient-based interventions should therefore be coupled with other interventions targeting weight control, such as exercise.

It is worth contrasting the results from Solon-Biet et al. (63) with the control/sham group from Rodriguez Paris et al. (20), because both measured the number of corpora lutea in the same strain. Solon-Biet et al. (63) reported that the number of corporal lutea was maximized in animals

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with elevated C intake, whereas the trend in the control group in the PCOS study was reversed. The other key difference was that the control animals in the PCOS study displayed more corpora lutea overall. A likely explanation for the difference is that Solon-Biet et al. (63) sampled animals at 15 months, whereas Rodriguez Paris et al. (20) sampled at 4 months. A possible hypothesis explaining the different results is that high-C, LP diets preserved some degree of reproductive function later in life by reducing reproductive investment earlier on.

Maternal effects. Freire et al. (78) examined the effects of maternal diet on offspring appetite, physiology, and metabolic health in C57Bl/6J mice. Mature female animals were put on either a HP or LP diet (14.7 kJ/g) for four weeks before being mated to males from a standard diet (AIN-93G). The intake and appetite for macronutrients in the offspring were then tested using choice and no-choice designs.

In a choice experiment offering both HP and LP foods from 3 to 8 weeks of age, offspring from HP dams consistently consumed more P than those from LP dams. There was no effect of maternal diet on C intake in the offspring. Offspring from the HP mothers also consumed more energy and displayed elevated body and fat mass. These effects were consistent across the sexes. This first experiment suggests that maternal diet affects the IT of the offspring, with a HP maternal diet setting a higher P target (78). In a second experiment, the offspring were subjected to a no-choice design with either a standard (AIN-93G) or a Western-style (high-F, high-sucrose at 17.3 kJ/g) diet. Animals on the Western diet displayed elevated energy intake and F mass and poorer glucose tolerance, regardless of maternal diet. However, on a standard diet, those animals from HP mothers displayed elevated energy intake and F mass. HP maternal diets also translated to poorer glucose tolerance, particularly among male offspring when exposed to a Western diet.

Overall, Freire et al.'s (78) results suggest that a HP maternal diet sets a higher P target in the offspring, which extends into adulthood. This high P target may predispose offspring to excess energy intake, weight gain, and metabolic disease via intensified P leverage in P-dilute food environments. These data may provide a mechanism to help explain the observation that HP infant formula feeding increases susceptibility to obesity in children and adolescents (79–81; see also 6).

Male reproductive function. Solon-Biet et al. (63) reported the effects of macronutrients on aged male reproductive function in the large GFN mouse study. They looked at testes, seminal vesicle, and ventral prostate mass, as well as plasma levels of testosterone. As in females, low energy intake was associated with poor outcomes. Increased P intake very strongly enhanced ventral prostate mass and plasma levels of testosterone, with weaker or nonexistent effects from the other macronutrients. However, seminal vesicle and testes mass were maximized with high intakes of energy where P and C were present at equal ratios (F had little effect). As was the case with females, the P content of the diet that maximizes reproductive function in aged male mice is likely to be higher than that maximizing lifespan (14, 63).

Crean et al. (33) used the GFN to examine the effects of dietary macronutrients on a much wider range of male reproductive traits than in Solon-Biet et al.'s (63) work, using C57Bl/6J males. They used an isocaloric design (14.7 kJ/g), with the % energy from P being 7–50%, C being 20–78%, and F being 15–60%. Across the 19-week study, the outcomes measured included morphological and endocrine parameters, basic sperm morphology and semen parameters, flow cytometry assessment of sperm viability and ROS production, testicular gene expression, and antioxidant activity of sperm and testes.

In this experiment, the effects of macronutrients on seminal vesicle mass largely aligned with those reported by Solon-Biet et al. (63), with equal and relatively high ratios of P and C maximizing the outcome (33). Crean et al. (33) did not detect the same effects of dietary P on circulating testosterone reported by Solon-Biet et al. (63). However, they did detect a positive effect of P

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on testicular expression of hydroxysteroid 17-β dehydrogenase 3, a precursor of testosterone. In terms of sperm, total motility and antioxidant capacity were maximized on LP diets with moderate or high F content and minimized on high P and low F. However, no statistically significant effects of macronutrients on sperm concentration, or on sperm viability, were reported. Higher P also led to decreased total antioxidant capacity, but this effect could not be attributed directly to elevated expression of antioxidant enzymes in the testes. The results from Crean et al.'s (33) study show that no single diet composition is likely to maximize male reproductive traits in mice simultaneously.

More recently, Crean et al. (82) reported a study using the GFN to look at male reproductive function. The data came from the experiment described above by Wali et al. (42), which sought to separate the effects of fructose and glucose at three different C:F ratios. Testes and seminal vesicle mass were quantified in the animals, and sperm from the epididymis were sampled and assessed for kinetic parameters. The study found generally that F acted additively with the ratio of fructose to glucose in the diet (82). Diet had relatively little effect on testes mass, a result that aligns with Solon-Biet et al.'s (63); the latter showed that testes mass is principally affected by substituting dietary P for C, whereas P was fixed in the current study. There was a minor effect of fructose:glucose ratio on seminal vesicle mass, whereby equal parts led to elevated total mass, an effect that was reflected in the body mass of the animals. Sperm concentration went up strongly with increasing fructose:glucose ratio, with the high-F diet additively increasing concentration. Parameters of sperm kinetics were not significantly affected by the C dimensions in this study, though motility fell slightly with increasing F content (63).

The pair of studies by Crean and colleagues clearly place the literature on Western diet patterns and male reproductive health into a new context. Many such studies in rodents have concluded that the high-F Western diet pattern results in poor male reproductive outcomes (83). The results from the two GFN studies, however, suggest that high F or fructose intakes are not necessarily detrimental and may in some cases improve motility and sperm concentration. Rather, the effects in the existing Western diet literature may be a secondary consequence of the high energy content of the diets and the weight gain it induces.

Paternal effects. The male mice in the P:C:F study reported by Crean et al. (33) were subsequently crossed with females on control diets. The offspring from the cross were then raised on a standard diet, isolating the effects of paternal diet. Both metabolic and behavioral phenotypes of the fathers and the offspring were assessed (34). Sex-specific effects of paternal diet were detected. The male offspring displayed a behavioral phenotype not present in females. Behavior was assessed by placing animals in an elevated plus maze, where the time spent in closed or open portions of the maze is measured. Spending more time in the closed space is often interpreted as an anxiety-like behavior (84, 85). The paternal diet had a clear effect on the amount of time male offspring spent in the closed maze, with an increasing P:C ratio driving less time in the closed arm. Incidentally, an identical phenotype was observed in the fathers themselves, where the effect of diet was even more pronounced (34). For female offspring, increasing paternal dietary F resulted in elevated subcutaneous fat, reduced brown adipose tissue, and a tendency toward elevated fasting glucose. After accounting for various cofactors that arose out of the experiment, such as variations in litter size and paternal food intake, additional metabolic responses were observed in both sexes, with elevated % body fat in females and increased AUC in males coinciding with high F intake in the fathers.

Microbiome

GFN studies in mice have helped tackle the complex association between diet and gut microbiome. Holmes et al. (86) reported on the effects of dietary macronutrients on microbiome composition

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in the animals from the large GFN experiment first reported by Solon-Biet et al. (14). Their analyses suggested that the major axis of effect was increasing total intake of P and C, whereas F had little to no effect (86). Low intakes were associated with the highest measures of alpha diversity (i.e., inverse Simpson index) at the class, family, and genus levels. Taxonomic richness showed no similar trend. The abundance of individual taxa could generally be classified as being positively or negatively correlated with increasing P and C intake, with most individual taxa having a negative correlation. The overall abundance of *Bacteroidetes* species was negatively correlated with increasing P and C, whereas *Firmicutes* showed the opposite trend. However, even within these phyla, individual taxa displayed opposing effects. Holmes et al. (86) concluded that dietary responses of microbiomes are best thought of within a trophic response guild framework, as opposed to a phylogenetic framework; that is, taxa should be classified as belonging to one of a small number of sets based on their common response to resource availability. They proposed that low intakes of P and C by the host promote a diverse array of taxa belonging to a guild that utilizes endogenous nitrogen. In contrast, higher intakes of P and C result in the proliferation of a guild that relies on host-ingested dietary nitrogen, possibly causing a dysbiotic relationship with the host.

Rodriguez Paris et al. (87) published a second study reporting on the effects of dietary macronutrients on rodent microbiome using the GFN. This experiment was isocaloric, and so could capture effects of dietary macronutrient composition but not absolute intake of macronutrients. They found that the total number of taxa present was unaffected by diet composition, though evenness and Shannon's diversity index were. Diets dominated by any single nutrient had the lowest levels of alpha diversity, whereas moderate compositions (e.g., P:C:F% = 28:38:38%) had maximal levels. Notably, however, in absolute terms the effects of diet on alpha diversity reported here are much smaller than those reported by Holmes et al. (86). This contrast is consistent with Holmes et al.'s conclusion that the biggest axis of effect on microbial alpha diversity is total energy, which was controlled by Rodriguez Paris et al. (87).

Before moving on to immune function, we want to touch on one last study in this section, which, rather than looking at the effects of diet on microbiome, looked at the reverse. Trevelline & Kohl (13) took germ-free mice and seeded their microbiomes using one of three donor species from different trophic levels: a carnivorous (*Onychomys torridus*), omnivorous (*Peromyscus leucopus*), and herbivorous (*Microtus monatus*) mouse species. The authors first confirmed that these different donors resulted in distinct microbial communities. They then evaluated the location of the P:C IT using a choice GFN design. Mice within all three groups consumed the same amount of P on average. However, mice seeded with the herbivorous microbiome consumed significantly less C. Mice within the herbivorous group also exhibited higher levels of circulating essential amino acids, including tryptophan, which was linked previously to dietary macronutrient preference (49), and had larger colons. The authors concluded that these results are consistent with the hypothesis that the microbiomes of herbivorous species directly affect their host's foraging behavior and physiological adaptations. However, robust support for this interesting hypothesis needs to come from a study that seeds germ-free mice with multiple donor species within each ecological category.

Immunity

The GFN has been applied to study the immune system in rodents in three contexts, all of which sought to separate the effects of P, C, and F. Le Couteur et al. (88) reported effects of macronutrient intake in the 15-month-old animals from Solon-Biet et al.'s (14) study. Lymphocytes from liver, spleen, Peyer's patches, mesenteric lymph nodes, and inguinal lymph nodes were quantified. Of all the compartments quantified, diet affected liver and spleen most profoundly, though correlated responses were seen in lymph nodes. P intake had the biggest effect, and the authors noted little

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effect coming from total energy. Increasing P intake drove a decrease in splenic CD4 T cells and an increase in CD8 (i.e., the CD4:CD8 ratio went down with elevated P). The number of naïve CD4 T cells in spleen also went down with increasing P, whereas the number of memory CD4 T cells went up. In liver, P intake positively affected the total number of B lymphocytes but negatively affected the number of natural killer cells. The effects of elevated P intake on the immune system observed here (e.g., loss of naïve T cells, decreased CD4:CD8 ratio, and increased B cells) are commonly seen in the immune systems of senescent and aged animals (89–92). Such immune senescence has previously been alleviated in mice by conventional CR (93, 94). The GFN results Le Couteur et al. (88) presented suggest that the immuno-protective effects of CR may be achievable using protein restriction in an ad libitum setting and likely are not due to a net reduction in calorie intake.

Tan et al. (95) studied the effects of dietary macronutrients on B cell development and function in younger C57Bl/6J male mice. Given that dietary macronutrients are the major sources of energetic substrate for immune cells, Tan et al. (95) initially used the GFN to test how diets varying in macronutrient content (P of 5–60% and C and F of 25–75%, fixed at 14.5 kJ/g) affected B cell differentiation. C content of the diet positively affected the proportion of B cells within the spleen, mesenteric lymph nodes, and Peyer's patches. The authors also separated the effects of glucose and fructose using a model similar to that of Wali et al. (42). Increasing dietary glucose led to B cell development via glycolysis and reduced mTOR-activated apoptosis of B cells (95).

Finally, Tan et al. (96) studied the effects of macronutrients on secretory immunoglobulin A (sIgA) using the same model described in Reference 95. sIgA plays a complex but fundamental role in mucosal homeostasis in the gut, both excluding pathogens and facilitating mutualism between the host and the gut microbiota (97). Furthermore, the presence of gut microbiota themselves play a causal role in sIgA production in rodents (98). Their analyses suggested that levels of luminal gut sIgA production were predominantly and positively affected by dietary P. Elevated sIgA levels were observed only after 5 weeks of high P feeding and were reversible following diet switch. The authors could trace the effects of dietary P on sIgA production to elevated rates of extracellular vesicle production by gut microbiota exhibiting a stress response. These extracellular vesicles activate Toll-like receptor 4 to increase epithelial expression of multiple pro-sIgA components and at least one transporter (96).

CONCLUSIONS

We have provided an overview of the effects of nutrition on rodent biology that have been uncovered using GFN designs. Coupling the GFN method to rodent models has led to new concepts in nutritional physiology (such as protein leverage and the role of FGF21 in protein appetite and responses) (6, 45) and is resolving key controversies in metabolism and organismal biology. Added to the various new insights described above, the application of GFN to the rodent model system has highlighted several areas for future research. First, most rodent GFN studies have been performed on C57Bl/6J mice, but alternative genetic strains may show different responses (e.g., 31, 99, 100). Assessing where this variation exists, and the causes, will provide an extra layer in our understanding of the mechanistic basis of nutrient-specific metabolic processes. Applying a 10+-diet GFN experiment to multiple genetic strains will be challenging, but experiments including selected combinations of diets from the existing literature are feasible. Second, we have identified that nutrients likely interact with life stage and physiological state to determine outcomes. A striking example here is the age-specific effect of nutrition on outcomes across the life course from growth and reproductive function to the age-specific risk of mortality (3); a life course study that samples animals longitudinally is needed to map these

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dynamics. Third, these age-specific effects are a specific example of the more general finding that dietary protein can induce pleiotropic trade-offs. As we have reviewed here, higher levels of dietary P may improve reproductive function, reduce excessive energy intake, and improve body composition. However, over the long term, these diets reduce insulin sensitivity and median lifespan. Resolutions to this protein paradox are likely to lie in using the GFN to titrate dietary P content as a function of life stage (e.g., 19), the amino acid composition of dietary P (e.g., exome matching; 101), and interactions between P and the ratio and quality of dietary C and F (42, 49, 55, 73).

Finally, it is worth reflecting on the wider value of rodent models of nutritional physiology to human nutrition. Thanks to its genetic similarities with humans (102, 103), small size, relatively short lifespan, and ease of breeding, the mouse has been the major mammalian model for more than a century (104). There are, however, dangers of overextending the model analogy. Nutrition-related examples of "physiological anthropomorphism" (104) include testing formulations that are beyond the realms of the model's natural environment, overinterpreting tests, and presuming that independently derived phenomena and their mechanisms are analogous (105). Having said that, when interpreted judiciously, results from rodents, coupled with other models, help to forge a "consilience of inductions" (106–108). The GFN is a powerful tool that, with cutting-edge application to nutritional epidemiology, is building a cross-taxa consilience around nutritional biology (e.g., 109–112). Furthermore, much of the knowledge that the GFN is generating has remained elusive, or in fact been confused, by simpler paradigms that manipulate a single nutritional dimension at a time.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding or financial holdings that might be perceived as affecting the objectivity of this review.

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